Structural Barriers to Geometric Dark Energy and the Perturbation-Transparency of Einstein-Cartan-Holst Gravity

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Abstract

We present a systematic catalog of structural barriers that prevent nonsingular bouncing cosmologies from generating late-time dark energy through first-principles mechanisms. Through 7 foundation studies and 17 research branches testing every viable theoretical route within the Einstein-Cartan-Holst (ECH) framework, we identify 14 independent barriers that close all standard mechanism classes. The barriers range from mass-coupling locks and topological-shift dualities to scale-separation arguments and perturbation-transparency theorems. Our central theoretical result is a perturbation-transparency theorem for minimal ECH gravity: for canonical scalar field matter, the spin density vanishes identically, torsion is algebraically zero at all perturbation orders, the Holst term reduces to the Pontryagin topological density, and the Barbero-Immirzi parameter becomes invisible in all scalar and tensor perturbation observables. This establishes ECH as a perturbation-transparent UV regulator—it resolves the Big Bang singularity through the spin-torsion modification of the Friedmann equation at Planck density, but leaves zero fingerprint on the primordial perturbations that survive into the CMB. We also document the systematic rejection of the hybrid dark-energy loophole: appending late-time dynamical-dark-energy freedom (e.g., CPL parametrization) to a bounce model was explored across 7 disguised forms and rejected on first-principles grounds, as it does not derive the dark-energy signal from the bounce mechanism. The observable science case for bouncing cosmology therefore rests entirely on generic, mechanism-independent predictions such as the matter-bounce non-Gaussianity $f_{\mathrm{NL}} = -35/8$.

1 Introduction

The Einstein-Cartan-Holst (ECH) formulation of gravity extends general relativity by allowing spacetime torsion, sourced by the spin density of matter [Holst, 1996]. In cosmological settings, the spin-torsion coupling produces a repulsive effective potential at Planck-scale densities, resolving the Big Bang singularity through a nonsingular bounce [**?**]. A natural question is whether this modified gravitational dynamics can also address other open problems in cosmology—particularly late-time cosmic acceleration (dark energy) and distinctive perturbation-level observables.

This paper presents the results of a systematic investigation of these questions, documenting both the negative structural results (14 barriers closing all standard dark-energy derivation routes) and the central positive theoretical finding (the perturbation-transparency of minimal ECH gravity).

2 The 14 Structural Barriers

We tested 7 foundation mechanism classes (Foundations A–G) and 7 additional observational channels (Branches H–O, plus the ECH perturbation gates) for the possibility of connecting the bounce to late-time dark energy or producing distinctive observable signatures. Each test yielded a named structural barrier.

#	Barrier	Source	Mechanism Blocked
1	Mass-Coupling Lock	Found. A	Propagating torsion as DE
2	Topological-Shift Duality	Found. B	Geometric pseudoscalar mass protection
3	Scalar-Tensor Universality	Found. C	Distinctive geometric content on FRW
4	Planck Suppression	Found. D	Disformal / connection coupling effects
5	Scale Separation	Found. E	Global vacuum integral coupling
6	Attractor-Sensitivity Dilemma	Found. F	Initial-condition transfer to DE
7	Parameter Immunity	Found. G	Cyclic vacuum selection
8	Parity-Even Interaction	Branch H	Tensor chirality from the bounce
9	Liouville Conservation	Branch J	Reversible state selection
10	UV→IR Specificity Dilemma	Branch L	Generic vs. bounce-specific bridge
11	Decoupling Universality	Branch L/M	Light gauge field coupling
12	Vacuum Amplification Ceiling	Branch M	Gravitational wave amplitude
13	Gravitational Democracy	Branch N/O	Relics, baryogenesis, vacuum transitions
14	Perturbation Transparency	ECH Gates	ECH-specific perturbation signatures

Table 1: The 14 structural barriers. Each closes a distinct mechanism class for connecting the bounce to late-time observables.

2.1 Barrier 1: Mass-Coupling Lock (Foundation A)

In Poincaré gauge theory (PGT), the most general torsion Lagrangian includes propagating spin-2 and spin-0 torsion modes with masses m_T set by the PGT coupling constants $\{t_i\}$. For these modes to act as dark energy, they would need to be ultralight ($m_T \sim H_0 \sim 10^{-33}$ eV) and couple to matter at cosmologically relevant strength.

The mass-coupling lock arises because the effective coupling is:

$$g_{\text{eff}} \sim \frac{1}{M_{\text{Pl}}\sqrt{|t_3|}} \tag{1}$$

where t_3 is the PGT coupling. For ultralight masses ($|t_3| \sim M_{\text{Pl}}^2/H_0^2$), the coupling becomes $g_{\text{eff}} \sim H_0/M_{\text{Pl}}^2 \sim 10^{-61}$—far too weak for any observable effect. To achieve $g_{\text{eff}} \sim 1$, one needs $|t_3| \sim 1/M_{\text{Pl}}^2$, giving $m_T \sim M_{\text{Pl}}$ (Planck-mass torsion, not dark energy).

The fine-tuning required to simultaneously achieve $m_T \sim H_0$ and $g_{\text{eff}} \sim \mathcal{O}(1)$ is:

$$\frac{m_T^2}{m_T^2|_{\text{natural}}} \sim \frac{H_0^2}{M_{\text{Pl}}^2} \sim 10^{-122} \tag{2}$$

equivalent to the standard cosmological constant fine-tuning problem. The barrier transfers rather than solves the fine-tuning.

Graviton loop corrections generate $\delta m_T^2 \sim M_{\text{Pl}}^2/(16\pi^2)$, requiring cancellation at the level of 1 in 10^{57}.

2.2 Barrier 2: Topological-Shift Duality (Foundation B)

Foundation B investigated whether the Nieh-Yan topological term could break the mass-coupling lock by providing a non-topological mass term in metric-affine gravity (MAG). In MAG, the Nieh-Yan term $\int T^a \wedge T_a - e^a \wedge e^b \wedge R_{ab}$ is indeed non-topological (unlike in standard EC theory).

However, a duality emerges: configurations that protect the pseudoscalar mass (through the topological structure) necessarily eliminate the geometric content (the field becomes a standard ALP after torsion elimination). Conversely, configurations that preserve geometric content cannot protect the mass. This *topological-shift duality* means:

$$\text{Mass protection} \iff \text{No geometric fingerprint} \tag{3}$$

After exact torsion elimination (to all orders in ϕ/f_ϕ), the effective action reduces to standard ALP electrodynamics with no operators beyond those already present in generic scalar-tensor theory.

2.3 Barrier 3: Scalar-Tensor Universality (Foundation C)

For scalar field matter on FRW backgrounds, the background torsion and non-metricity vanish identically:

$$T_0 = Q_0 = 0 \quad \text{(exact on FRW with scalar matter)} \tag{4}$$

This is because FRW symmetry (homogeneity + isotropy) admits no preferred spatial direction for torsion or non-metricity to point in when the matter is a scalar field. Environmental mass mechanisms (chameleon-like effects from torsion) therefore reduce exactly to standard scalar-tensor theory on cosmological backgrounds, with no distinctive geometric fingerprint.

2.4 Barrier 4: Planck Suppression (Foundation D)

Disformal couplings (terms involving $\partial_\mu\phi\,\partial_\nu\phi$ in the effective metric) could in principle produce distinctive signatures. However, in the connection-coupling framework of ECH, each interaction vertex carries only one $\partial\phi$ factor (from the single derivative in the torsion-scalar coupling), while disformal effects require two. The resulting operators are suppressed by:

$$\frac{\text{Disformal effect}}{\text{Conformal effect}} \sim \frac{k^2}{M_{\text{Pl}}^2} \sim 10^{-122} \tag{5}$$

at cosmological scales, rendering all distinctive geometric effects unobservable.

2.5 Barrier 5: Scale Separation (Foundation E)

Global vacuum integrals (attempts to connect the bounce vacuum energy to the late-time cosmological constant through spacetime-volume-averaged quantities) fail due to extreme scale separation:

$$\frac{V_4^{\text{bounce}}}{V_4^{\text{total}}} \sim \frac{t_{\text{bounce}}^4}{t_0^4} \sim \frac{t_{\text{Pl}}^4}{(10^{17}\,\text{s})^4} \sim 10^{-244} \tag{6}$$

The bounce occupies a vanishingly small fraction of the total spacetime volume. No averaging procedure can amplify the bounce-scale vacuum energy to affect late-time dynamics.

2.6 Barrier 6: Attractor-Sensitivity Dilemma (Foundation F)

Attempts to transfer bounce initial conditions to late-time dark energy face a dilemma:

- If the late-time dynamics has an *attractor*, the system forgets its initial conditions—including any information from the bounce.
- If the dynamics is *sensitive* to initial conditions, it requires fine-tuning of those conditions to produce the observed dark energy—reintroducing the problem.

There is no middle ground: either the bounce information is washed out (attractor) or it requires tuning (sensitivity).

2.7 Barrier 7: Parameter Immunity (Foundation G)

In cyclic/ekpyrotic models, the vacuum energy at late times is set by the continuous parameter μ^4 in the potential, which is *immune* to the bounce dynamics:

$$\Lambda_{\text{eff}} = \mu^4 + \mathcal{O}(e^{-M_{\text{Pl}}/\sigma}) \approx \mu^4 \tag{7}$$

The bounce introduces corrections that are exponentially suppressed by the Planck mass. The vacuum energy is a free parameter, not a prediction.

2.8 Barrier 8: Parity-Even Interaction (Branch H)

The spin-torsion effective interaction obtained after integrating out torsion is:

$$\mathcal{L}_{\text{eff}} \supset \frac{3}{16}\frac{(J^5_\mu)^2}{M_{\text{Pl}}^2} \tag{8}$$

where $J^5_\mu = \bar{\psi}\gamma^\mu\gamma^5\psi$ is the axial fermion current. Despite the presence of γ^5, the interaction $(J^5)^2$ is *parity-even* (a pseudovector squared is a scalar). This means:

- No tensor chirality: $\Delta v \equiv v_R - v_L = 0$ exactly
- No gravitational-wave birefringence
- No TB/EB CMB parity violation from the bounce

FRW isotropy further kills all spatial parity-odd backgrounds.

2.9 Barriers 9–11 (Branches J, L)

Barrier 9 (Liouville Conservation): Phase-space conservation (Liouville's theorem) prevents the bounce from irreversibly selecting a preferred state. Any state contraction would violate unitarity.

Barrier 10 (UV→IR Specificity Dilemma): Extensions bridging UV (bounce-scale) to IR (cosmological-scale) physics face a dilemma: generic bridges produce effects indistinguishable from standard scalar-tensor theory, while specific bridges require tuning.

Barrier 11 (Decoupling Universality): Light gauge fields decouple from Planck-scale torsion modes with universal efficiency, preventing any bounce-scale information from reaching the gauge sector.

2.10 Barrier 12: Vacuum Amplification Ceiling (Branch M)

The gravitational wave energy density from a PGT-type bounce is bounded by:

$$\Omega_{\text{GW}}(f) \propto \left(\frac{H_{\text{bounce}}}{M_{\text{Pl}}}\right)^2 \propto \left(\frac{f_{\text{bounce}}}{f_{\text{Pl}}}\right)^4 \tag{9}$$

For any sub-Planckian bounce, Ω_{GW} is catastrophically small. The characteristic frequency is $f_{\text{bounce}} \sim 10^9$–$10^{10}$ Hz (GHz), with zero overlap with any planned detector (LIGO at 10–10^3 Hz, LISA at 10^{-4}–10^{-1} Hz). The detector gap is at least 10^{17} in amplitude.

2.11 Barrier 13: Gravitational Democracy (Branches N/O)

Torsion-mediated baryogenesis or relic production fails because torsion couples democratically to all fermion species ($\sim 1\%$ torsion contribution per species), with no mechanism to preferentially produce baryons over antibaryons or specific dark matter candidates. Vacuum transitions at the bounce are blocked by the bounce-vacuum decoupling: the bounce is too brief and too symmetric to trigger irreversible phase transitions.

2.12 Barrier 14: Perturbation Transparency

See Section 3 for the full proof. In summary: for canonical scalar field matter, torsion vanishes identically at all perturbation orders, rendering the Holst term topological and the Barbero-Immirzi parameter invisible.

3 The Perturbation-Transparency Theorem

3.1 Statement

In minimal Einstein-Cartan-Holst gravity with canonical scalar field matter, the Holst term is dynamically inert for both scalar and tensor perturbations at all orders. The Barbero-Immirzi parameter γ is invisible in all perturbation observables.

3.2 Proof (Scalar Sector)

The argument proceeds in five steps:

1. **Zero spin density.** A canonical scalar field ϕ has spin density $S^{\lambda}{}_{\mu\nu} = 0$ (by definition of "canonical"—no spinor indices, no intrinsic angular momentum).

2. **Zero torsion.** In Einstein-Cartan theory, the torsion tensor is algebraically determined by the spin density: $T^{\lambda}{}_{\mu\nu} = 8\pi G\,(S^{\lambda}{}_{\mu\nu} + \delta^{\lambda}_{[\mu} S_{\nu]} + \ldots)$. With $S = 0$, we have $T^{\lambda}{}_{\mu\nu} = 0$ at all perturbation orders.

3. **Connection reduces to Levi-Civita.** With $T = 0$, the full connection $\Gamma^{\lambda}{}_{\mu\nu} = \mathring{\Gamma}^{\lambda}{}_{\mu\nu}$ (Christoffel symbols only).

4. **Holst term becomes topological.** The Holst term $\frac{1}{2\gamma} \int \epsilon^{IJ}{}_{KL}\, e^K \wedge e^L \wedge F_{IJ}$ evaluated with the Levi-Civita connection reduces to the Pontryagin density $\int \sqrt{-g}\, \epsilon^{\mu\nu\rho\sigma} R_{\mu\nu\rho\sigma}\, d^4x$, which is a total derivative in 4D.

5. **No equations of motion.** A total derivative contributes zero to the variational equations at all orders in perturbation theory.

3.3 Extension to Tensor Sector

The same five-step argument applies to tensor perturbations. With $T = 0$, the gravitational action reduces to the Einstein-Hilbert form plus a topological term. The tensor perturbation equation is:

$$h''_{ij} + 2\mathcal{H}h'_{ij} + k^2 h_{ij} = 0 \tag{10}$$

with no parity-dependent modifications. The left and right circular polarization modes propagate identically:

$$v_R(k,\eta) = v_L(k,\eta) \quad \Rightarrow \quad \Delta v \equiv v_R - v_L = 0 \quad \text{(exact)} \tag{11}$$

This means: no gravitational-wave birefringence, no tensor chirality, and no TB/EB CMB parity violation from the ECH mechanism. This result was independently confirmed by the Branch H parity-tensor analysis (Barrier 8), which proved that the spin-torsion effective interaction $(J^5)^2$ is parity-even.

3.4 Explicit Verification: The Holst Term in Perturbation Theory

To make the argument fully explicit, consider the ECH action:

$$S_{\text{ECH}} = \frac{M_{\text{Pl}}^2}{2} \int d^4x \sqrt{-g} \left[R(\Gamma) + \frac{1}{\gamma} \widetilde{R}(\Gamma) \right] + S_{\text{matter}}[\phi, g] \tag{12}$$

where $\widetilde{R} = \frac{1}{2}\epsilon^{\mu\nu\rho\sigma} R_{\mu\nu\rho\sigma}$ is the Holst dual of the Ricci scalar, and Γ is the full (torsionful) connection.

Expanding the metric to second order ($g_{\mu\nu} = \bar{g}_{\mu\nu} + \delta g_{\mu\nu}^{(1)} + \delta g_{\mu\nu}^{(2)} + \ldots$) and the scalar field ($\phi = \bar{\phi} + \delta\phi^{(1)} + \delta\phi^{(2)} + \ldots$):

At *each perturbation order*, the torsion equation of motion gives $T = 0$ (because S_{matter} for a canonical scalar has zero spin density at every order). Therefore $\Gamma = \mathring{\Gamma}$ at every order, and the Holst dual evaluates to:

$$\widetilde{R}(\mathring{\Gamma}) = \frac{1}{2}\epsilon^{\mu\nu\rho\sigma} R_{\mu\nu\rho\sigma}(\mathring{\Gamma}) = \text{total derivative} \tag{13}$$

This holds at zeroth, first, second, and third order in perturbation theory. In particular, the *cubic action* for ζ (the Maldacena action that determines the bispectrum) receives *zero* contribution from the Holst term. The bispectrum is therefore identical to the standard GR result.

3.5 What Would Break the Transparency

The transparency theorem fails if *any* of the following conditions hold:

1. **Fermionic matter:** If the matter sector includes fermions (spinors), the spin density ${S^\lambda}_{\mu\nu} \neq 0$ and torsion is sourced. The Holst term then contributes dynamically. However, fermionic contributions to torsion are Planck-suppressed at cosmological densities and negligible during the matter-contraction phase relevant for the bispectrum.

2. **Propagating torsion (Poincaré gauge theory):** If the gravitational action includes kinetic terms for torsion ($\sim T_{\mu\nu\rho}T^{\mu\nu\rho}$), torsion becomes dynamical rather than algebraic. The Holst term then generates non-trivial dynamics. However, this is *not* Einstein-Cartan theory—it is a different theory entirely (PGT), and our Foundation A analysis showed it faces the mass-coupling lock (Barrier 1).

3. **Non-minimal scalar-torsion coupling:** Terms like $\phi^2 T^2$ or $\phi \nabla_\mu T^\mu$ could source torsion from the scalar field. However, these are ad hoc additions to the minimal ECH action, and our Foundation C analysis showed they reduce to standard scalar-tensor theory on FRW backgrounds (Barrier 3).

4. **Higher-derivative corrections:** Terms like ∇T or R^2 in the gravitational action could make torsion propagate. These are UV corrections expected at the Planck scale and are negligible during the matter-contraction phase ($\rho \ll M_{\text{Pl}}^4$).

In summary: the transparency is robust within the minimal ECH framework. Breaking it requires going beyond minimal ECH, and all such extensions have been tested and closed by Barriers 1–4.

3.6 Implications

ECH gravity is a *perturbation-transparent UV regulator*: it resolves the singularity at Planck density through the ρ^2 modification of the Friedmann equation, but leaves zero fingerprint on the primordial perturbations. The Barbero-Immirzi parameter γ, which sets the bounce density

ρ_{crit} and plays a central role in loop quantum gravity, is completely invisible in all scalar and tensor perturbation observables computed with canonical scalar field matter.

This result has a positive interpretation: the observable predictions of a matter bounce (principally $f_{\text{NL}} = -35/8$) are *mechanism-independent.* They do not depend on whether the bounce is realized by ECH, LQC, or any other UV completion. This makes the predictions *more robust*, not less—they cannot be invalidated by changes to the bounce mechanism.

4 The Hybrid Dark-Energy Loophole

We considered the phenomenological route of appending late-time dynamical-dark-energy freedom (e.g., CPL $w_0 w_a$ parametrization) to the bounce model. This was explored across 7 disguised forms:

1. Direct $w_0 w_a$ addition to the background
2. Quintessence scalar with bounce-motivated initial conditions
3. Curvaton-derived late-time potential
4. Vacuum energy from cyclic boundary conditions
5. Torsion-induced effective $w(z)$
6. Holst-term residual as effective DE
7. ALP rolling as late-time acceleration

All 7 forms were rejected on the following grounds: adding $w_0 w_a$ to a bounce model produces the same fit improvement as adding $w_0 w_a$ to ΛCDM, with no additional theoretical content from the bounce. None of the 236,622 MCMC posterior samples in this program used w_0 or w_a as free parameters. The loophole was explored theoretically but never implemented computationally, because its implementation would not constitute first-principles evidence for bounce cosmology.

5 Consequences for the Observable Program

The 14 barriers, together with the perturbation-transparency theorem, establish that:

- ECH provides the *background* for the bounce (singularity resolution at ρ_{crit})
- ECH does *not* provide distinctive perturbation-level observables
- The observable science case for bouncing cosmology rests on *generic*, mechanism-independent predictions
- The strongest such prediction is $f_{\text{NL}}^{\text{local}} = -35/8$ [Cai et al., 2009], testable by SPHEREx at 4–6σ significance [Heinrich et al., 2023]

This is not a failure of the framework—it is a sharp scientific conclusion. The perturbation-transparency of ECH means that the observable predictions of a matter bounce are *more robust*, not less: they do not depend on the choice of bounce mechanism (ECH, LQC, or other UV completion).

6 Conclusion

We have presented a systematic catalog of 14 structural barriers that close all standard routes from nonsingular bounces to late-time dark energy or distinctive perturbation-level observables within the ECH framework. The central result is the perturbation-transparency theorem: minimal ECH is dynamically inert for scalar and tensor perturbations when the matter content is a canonical scalar field. The Holst term reduces to a topological invariant, and the Barbero-Immirzi parameter vanishes from all perturbation equations. The hybrid dark-energy loophole (appending phenomenological $w_0 w_a$ freedom) was explored across 7 forms and rejected as non-derivational. The observable program for bouncing cosmology therefore targets generic, mechanism-independent predictions—principally $f_{\rm NL} = -35/8$—rather than framework-specific signatures.

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